Supplementary Material for SeDa

A System Interface

Figure 1 shows the visualization interface of SeDa. It adopts a three-tier layout: the top section summarizes the queried topic with representative datasets, the middle section presents multi-entity cards for institution and enterprise exploration, and the bottom section provides related dataset cards for direct access to source platforms.

B Topic Annotation

This section provides additional analysis of the topic annotation module. We first present a case study to illustrate how SeDa recalls candidate tags and refines them. We further compare the generated tags with platform-provided annotations on representative datasets to evaluate the quality of the topic annotations.

B.1 Case Study

Figure 2 shows an example of tag graph construction and refinement for an urban road perception dataset. Candidate tags are recalled by multiple paths, including D2T, T2T, and D2D2T. After LLM-based refinement, the off-topic tag *Adverse Weather* is removed, while *Pedestrian Detection* and *Automatic Driving* are selected as final tags. The weakly related tag *Traffic Perception* is retained for auxiliary exploration rather than selected as a final annotation.

B.2 Topic Quality

To qualitatively evaluate our topic tag generation, we compare our tags with those provided by HuggingFace on representative datasets, as shown in Table 1. While HuggingFace typically provides generic tags such as `audio classification` or `image classification`, SeDa can identify more fine-grained and domain-specific topics such as `traditional Chinese instruments` and `tree species classification`. These results indicate that platform-provided tags often omit key contextual or culturally specific information, whereas our approach can capture such nuanced semantics and enhance the interpretability of dataset metadata.

C Deduplication Implementation Details and Examples

This section provides a concise yet comprehensive description of the three-stage deduplication pipeline implemented in SeDa.

In the first stage, explicit identifier matching is used to detect cross-platform replicas by aligning normalized dataset names and canonicalized URLs. Next, to avoid the quadratic complexity of global pairwise comparison, a hash-based blocking strategy is employed to partition the candidate space. Dataset titles and descriptions are concatenated into a unified textual representation and converted into text signatures. A hybrid blocking approach combining SimHash and Locality-Sensitive Hashing (LSH) is adopted: SimHash captures character-level similarity, while LSH preserves cosine similarity under TF–IDF vectorization. Only records that fall within the same hash bucket are considered potential duplicates and forwarded to the subsequent semantic matching stage, substantially reducing the comparison cost.

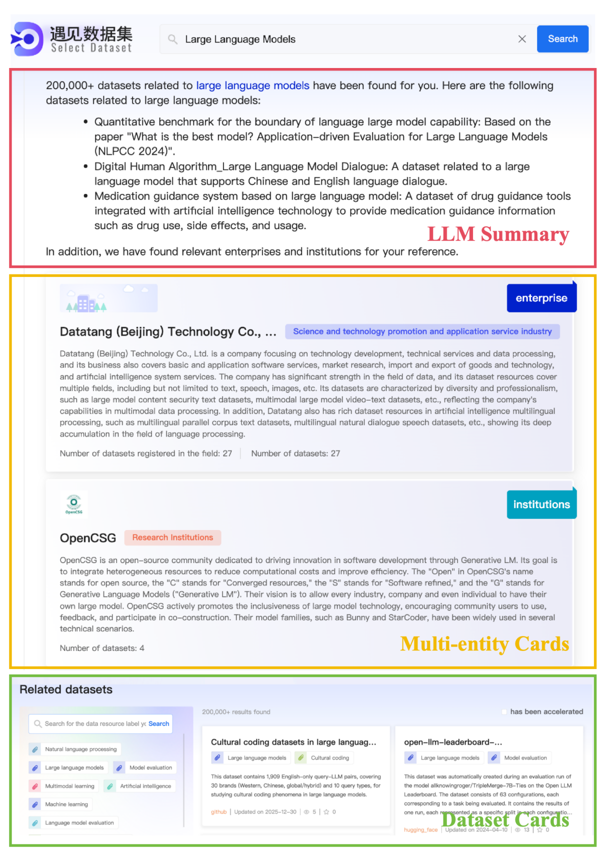

Figure 1: Navigation module visualization.

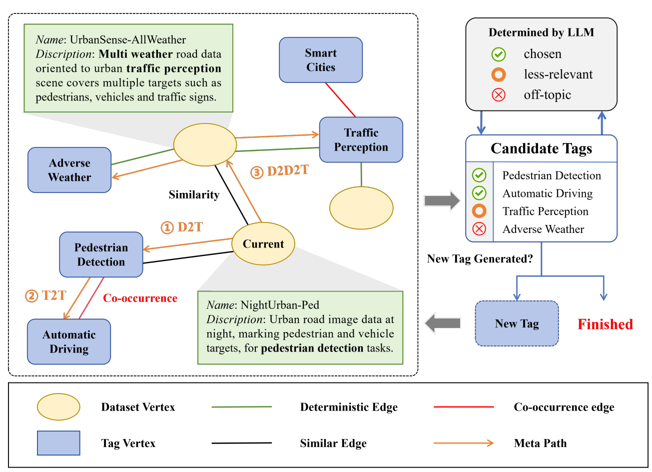

Figure 2: Tag graph construction.

In the semantic similarity assessment stage, `dataset_name` and `dataset_desc` are concatenated to form a unified text string for each record and encoded into an embedding using the sentence transformer `all-MiniLM-L6-v2`. Cosine similarity between dataset embeddings is computed, and record pairs whose similarity exceeds

Table 1: Comparison between our generated topic tags and HuggingFace annotations.

Dataset	Our Tags	HuggingFace Task	HuggingFace Tag	Description (excerpt)
CTI5	traditional Chinese instruments, music classification	audio classification	music, art	Recordings of Chinese traditional musical instruments for timbre recognition.
Rico	mobile application, user interface design	others	graphic-design	Large-scale dataset of mobile app screenshots and UI metadata.
TreeSatAI-T	tree species classification, temporal series analysis	image classification	earth-observation, remote-sensing	TreeSatAI time series dataset for tree species classification.
batman2	traditional Chinese medicine, drug discovery	others	biology, drug discovery	Molecular structures derived from traditional Chinese medicine for drug discovery.
clinc_oos	intent classification, out-of-scope query detection	text classification	intent classification	Dataset for out-of-scope intent detection in dialogue systems.

Table 2: Example of duplicate records detected by semantic similarity where explicit identifier matching fails.

Field	Record 1	Record 2
dataset_name	Tiny_Imagenet	Tiny ImageNet
dataset_desc	Tiny Imagenet has 200 Classes, each class has 500 traininig images, 50 Validation Images and 50 test images. Label Classes and Bounding Boxes are provided.	Tiny ImageNet contains 100000 images of 200 classes (500 for each class) downsized to 64×64 colored images. Each class has 500 training images, 50 validation images and 50 test images.
dataset_url	https://figshare.com/articles/dataset/Tiny_Imagenet	https://paperswithcode.com/dataset/tiny-imagenet

a predefined threshold are considered duplicates and merged. Semantic similarity search within each hash bucket is implemented using an approximate nearest neighbor index, with FAISS employed to ensure scalable and efficient retrieval.

All detected duplicate relations are organized into a deduplication graph, where each connected component corresponds to a unique dataset entity. Within each component, `SeDa` selects a canonical record to represent the entity using a rule-based strategy that accounts for metadata completeness, source reliability, and descriptive richness. Authoritative sources with more comprehensive metadata are preferred, while auxiliary information such as alternative URLs and attributes from other records is retained to preserve provenance traceability.

Table 2 presents a representative example illustrating the necessity of semantic similarity matching. The two records correspond to the `Tiny ImageNet` dataset but originate from different platforms with distinct URLs and slightly different naming conventions. Although explicit identifier matching and hash-based blocking alone are insufficient to resolve this duplication, semantic similarity matching correctly identifies the two records as duplicates. In this example, the cosine similarity between the two semantic representations reaches 0.89, exceeding the similarity threshold used in our system (0.85), thereby triggering a correct merge. This example demonstrates how the full pipeline complements the high-level deduplication design described in the main paper by resolving subtle cross-platform redundancy.

D Adaptive Site-level Dead-link Detection

This section presents validation and implementation details of our dead-link detection. Algorithm 1 outlines the site-level detection procedure, and Table 3 summarizes inspection results across representative platforms with varying scales at two system stages.

The first inspection round was conducted on June 25, 2025, shortly after repository initialization, when large-scale multi-source integration was still ongoing. The second round took place on September 20, 2025, after major data cleaning and consolidation had completed and the system reached a relatively stable state. Across all examined sources, alive rates exhibit consistent improvements in the second round, approaching near-complete accessibility, which confirms the effectiveness of the proposed monitoring mechanism.

Table 3: Two-round site-level sampling and alive-rate results. (S_1, S_2: sample size; A_1, A_2: alive rates.)

Site	N_s	S_1	A_1	S_2	A_2
DataCite	2,103,855	2,598	0.89	2,696	**0.99**
data.europa	1,761,468	2,564	0.88	2,654	**1.0**
Kaggle	359,561	1,148	0.96	1,199	**1.0**
HuggingFace	193,227	858	0.85	879	**0.98**
data.gov	71,424	531	0.86	534	**1.0**
arXiv	17,743	272	0.90	266	**1.0**
OpenData	7,758	180	0.74	176	**0.99**
Papers with Code	7,217	175	0.94	169	**1.0**
Aliyun Tianchi	4,170	132	0.96	129	**1.0**
Shanghai Data Open	1,021	59	0.79	63.9	**0.98**

E Prompt Templates Used in the System Pipeline

To provide a complete view of the system workflow, this section summarizes the prompt templates used across different stages of the pipeline. Each subsection corresponds to a specific system module

Algorithm 1 Adaptive Site-Level Dead-Link Detection

Require: Site set $\mathcal{S}$ with statistics $\{N_s, \sigma_s^2, \Delta N_s\}$, total budget K_{total}, and threshold $\tau = 0.9$
Ensure: Alive-site list $\mathcal{S}_{\text{alive}}$

(Weighting)
for $s \in \mathcal{S}$ **do**
  $w_s \leftarrow \log(1 + N_s)\ (1 + \lambda_1 \sigma_s^2)\left(1 + \lambda_2 \frac{\Delta N_s}{N_s + \varepsilon}\right)$
end for
(Budget Allocation)
for $s \in \mathcal{S}$ **do**
  $k_s^\star \leftarrow \dfrac{w_s}{\sum_{s' \in \mathcal{S}} w_{s'}} K_{\text{total}}$
  $k_s \leftarrow \min\big(k_{\max},\ \max(k_{\min},\ \text{round}(k_s^\star))\big)$
end for
(Sampling and Checking)
for $s \in \mathcal{S}$ **do**
  Randomly sample k_s URLs from site s
  Perform HTTP checks and compute $\text{AliveRate}(s)$
  if $\text{AliveRate}(s) < \tau$ **then**
    Mark s as **DEAD**, hide all datasets of s, and schedule rechecking
  else
    Mark s as **ALIVE**
  end if
end for
  return $\mathcal{S}_{\text{alive}}$

and documents the associated prompt template, clarifying how textual inputs and intermediate signals are structured for interaction with large language models.

E.1 Dataset Extraction Prompt

This subsection presents the unified dataset-centric extraction prompt introduced in Section 4. In general, source adapters indicate the input text type and highlight platform-specific cues for extraction. For GitHub README text, the adapter focuses on dataset-centric descriptions while filtering incidental mentions. For arXiv titles and abstracts, it emphasizes signals of dataset introduction or release, prioritizing dataset naming, construction methodology, scale, and author or institutional cues when available. For HuggingFace dataset descriptions, it guides the extraction of overall dataset descriptions, content composition, split information, and access or usage instructions.

E.2 Relevance Evaluation Prompt

This subsection presents the prompt used for LLM-based relevance evaluation in the ablation study. Given a user query, a selected subtopic, and two sets of retrieved results, the prompt guides the LLM to first assess each dataset item with fine-grained relevance scores and concise justifications, and then compare the overall quality of the two result lists. This step-by-step scoring process encourages the model to make more grounded judgments based on subtopic alignment, topic consistency, and semantic relevance.

E.3 Topic Refinement Prompt

This subsection presents the prompt used for dataset-level topic tag refinement. Given a dataset name, its textual description, and a set of candidate tags recalled from the tag graph, the prompt instructs the LLM to select exactly two representative tags as the final annotation.

Unified Prompt for Dataset Extraction

System
You are a dataset analysis expert specializing in extracting dataset-related information from unstructured textual records originating from heterogeneous sources. You will be provided with a piece of text content. Please carefully analyze the given text, and perform dataset-centric information extraction according to the instructions below.
Source Adapter
`{source_hint}`
Requirements

- Identify explicit or implicit dataset information in the text.
- If dataset information is present, extract and summarize its core characteristics, including the dataset name, overall description, and access or acquisition method.
- If no meaningful description can be obtained, leave the corresponding fields as empty strings.

Response Format

```
{
  "is_dataset": "Yes/No/Uncertain",
  "dataset_name": "",
  "dataset_desc": "",
  "dataset_url": "",
  "analysis": ""
}
```

Notes

- Do not access or rely on any external links in the text.
- All extracted information must be based on the provided content.
- The output should be concise and written in an academic style appropriate for dataset documentation.
- Do not include any comments, explanations, or formatting outside the specified JSON structure.

Input

- Text Content: `{record_text}`

When the candidate tags are insufficient to characterize the dataset, the LLM is allowed to generate new topic tags and explicitly mark them in the output. It also requires the model to distinguish weakly related tags from off-topic ones, so that noisy candidates can be filtered while potentially useful auxiliary topics are still preserved for exploration.

E.4 Summarization Prompt

This subsection presents the prompt used for query-oriented dataset summarization. Given a user query and a set of retrieved dataset records, the prompt guides the LLM to generate a concise and neutral summary that highlights representative datasets and their relevance to the query. When provider information is available, the prompt also allows the summary to mention related institutions, enterprises, or platforms, thereby complementing dataset-level retrieval with entity-level context.

Prompt for Topic Filtering Evaluation

System
You are a professional dataset relevance evaluation expert. The user is searching for datasets related to the topic "{query_text}", with a specific subtopic "{tag_meaning}". Your task is to assess how well two sets of retrieved results match the user's true needs. Result A and Result B are produced by two different retrieval strategies. Please remain neutral and base your judgment solely on dataset relevance and quality.

Evaluation Criteria

- **Subtopic Alignment**: The core content of the dataset should correspond to "{tag_meaning}", rather than merely mentioning related keywords.
- **Topic Consistency**: The dataset should align with "{query_text}", rather than belonging to cross-domain or weakly related scenarios.
- **Semantic Relevance**: The actual data content, application context, or annotation targets should closely match the user's intent, avoiding overly generic or ambiguous matches.

Evaluation Procedure
Step 1: Item-Level Scoring
Independently score each item in Result A and Result B using integer values:

- 3 (Perfect): Clearly satisfies "{tag_meaning}".
- 2 (Relevant): Generally relevant, but with minor deviations, vague descriptions, or limited noise.
- 1 (Weak): Mainly keyword-level matches and unlikely to satisfy real user needs.
- 0 (Irrelevant): Largely unrelated to the user intent, or clearly not a dataset (e.g., papers, models, tools, or code repositories).

For each item, provide one extremely concise justification.

Step 2: List-Level Aggregation
Compute:

- score_A: the average score of all items in Result A
- score_B: the average score of all items in Result B

Determine the winner:

- The result set with the higher score wins.If the two scores are equal, output "Tie".

Output Format (JSON only)

```
{
  "items_A": [
    {"rank": 1, "score": 0-3, "reason": "..."},
    {"rank": 2, "score": 0-3, "reason": "..."},
    ...
  ],
  "items_B": [
    {"rank": 1, "score": 0-3, "reason": "..."},
    {"rank": 2, "score": 0-3, "reason": "..."},
    ...
  ],
  "score_A": <float>,
  "score_B": <float>,
  "winner": "Result A" | "Result B" | "Tie",
  "reason": "A Brief explain"
}
```

Retrieved Results

- Result A: `{result_a}`
- Result B: `{result_b}`

Prompt for Topic Tag Refinement

System
You are a dataset annotation specialist responsible for assigning standardized topic tags to datasets. You will be provided with a dataset name, a textual description, and a set of candidate topic tags, and your task is to refine and finalize the dataset-level topic annotation.
Requirements

- Select **exactly two** representative topic tags that best characterize the dataset at the dataset level.
- Generate new tags only when the provided candidate tags are insufficient to produce two suitable and representative tags.
- Classify all remaining candidate tags as either weakly related or discarded based on their semantic relevance.

Response Format Response Format

```
{
  "selected": [
    {"tag": string, "is_new": boolean, "reason": string},
    {"tag": string, "is_new": boolean, "reason": string}
  ],
  "weakly_related": [string],
  "discarded": [string]
}
```

Notes

- `selected` must contain exactly two tags.
- New tags must appear in `new_tags`.
- Output JSON only.

Input

- Dataset Name: `{dataset_name}`
- Dataset Description: `{dataset_description}`
- Candidate Tags: `{candidate_tags}`

Prompt for Query Result Summarization

System
You are a dataset summarization specialist responsible for producing concise and coherent summaries of datasets in response to user queries.
Requirements

- Summarize the datasets relevant to the given user query in a neutral and informative manner.
- Dataset descriptions should be paraphrased for clarity and conciseness, rather than copied verbatim from the input.
- If provider information (e.g., institutions, enterprises, or platforms) is available, include a brief description; otherwise, omit this part.

Response Format
A single textual summary introducing the relevant datasets and, when applicable, the associated provider information.
Notes

- The response should contain only the summarized textual content.
- No additional explanations, markup, or formatting should be included.

Example
User Query: Datasets related to autonomous driving
Dataset Records:
KITTI Vision Benchmark Suite: A widely used autonomous driving dataset supporting tasks such as object detection, 3D tracking, stereo vision, and road segmentation across diverse driving scenes and conditions.
Cityscapes: An urban street-scene dataset designed for road and scene understanding, with images collected from multiple cities and seasons.
Provider Information:
DeepScenario: A research organization focused on autonomous driving scenario generation to enhance perception and decision-making capabilities.
Output: Representative autonomous driving datasets include the KITTI Vision Benchmark Suite for core driving perception tasks and Cityscapes for urban street-scene understanding. In addition, research organizations such as DeepScenario contribute complementary resources and expertise for autonomous driving research.
Input

- User Query: `{user_query}`
- Dataset Records: `{dataset_records}`
- Provider Information (optional): `{provider_info}`